0/1 Knapsack (medium)

Introduction

Given the weights and profits of ‘N’ items, we are asked to put these items in a knapsack with a capacity ‘C.’ The goal is to get the maximum profit out of the knapsack items. Each item can only be selected once, as we don’t have multiple quantities of any item.

Let’s take Merry’s example, who wants to carry some fruits in the knapsack to get maximum profit. Here are the weights and profits of the fruits:

Items: { Apple, Orange, Banana, Melon }

Weights: { 2, 3, 1, 4 }

Profits: { 4, 5, 3, 7 }

Knapsack capacity: 5

Let’s try to put various combinations of fruits in the knapsack, such that their total weight is not more than 5:

Apple + Orange (total weight 5) => 9 profit

Apple + Banana (total weight 3) => 7 profit

Orange + Banana (total weight 4) => 8 profit

Banana + Melon (total weight 5) => 10 profit

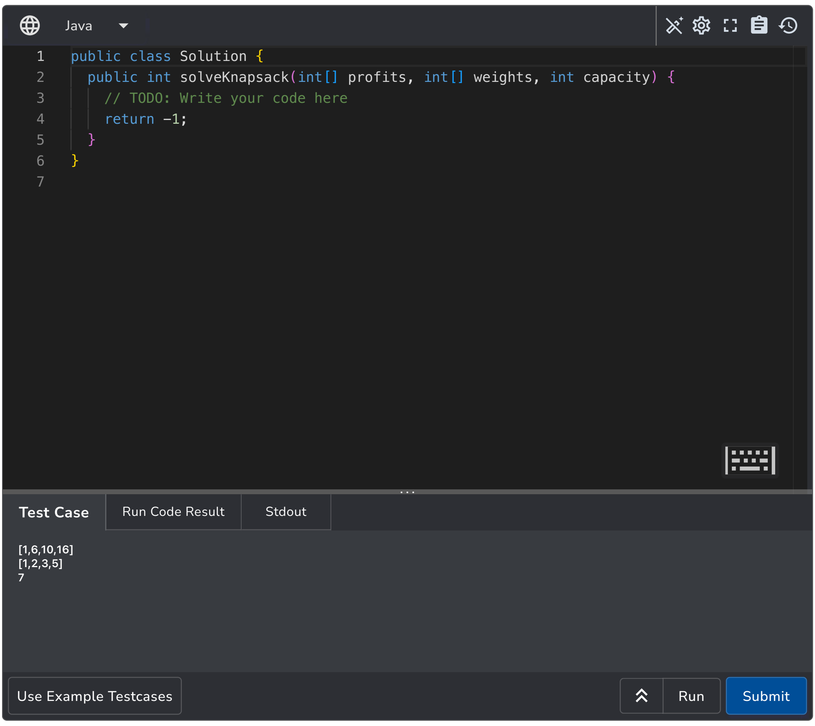
This shows that **Banana + Melon** is the best combination as it gives us the maximum profit, and the total weight does not exceed the capacity.

Problem Statement

Given two integer arrays to represent weights and profits of ‘N’ items, we need to find a subset of these items which will give us maximum profit such that their cumulative weight is not more than a given number ‘C.’ Each item can only be selected once, which means either we put an item in the knapsack or we skip it.

Try it yourself

Try solving this question here:



← Previous
Introduction To 0/1 Knapsack Pattern

Next →
Solution: 0/1 Knapsack

Mark as Completed